

Computation Resource Allocation and Task Assignment Optimization in Vehicular Fog Computing: A Contract-Matching Approach

Zhenyu Zhou^{1b}, Senior Member, IEEE, Pengju Liu^{1b}, Junhao Feng^{1b}, Yan Zhang^{1b}, Senior Member, IEEE, Shahid Mumtaz^{1b}, Senior Member, IEEE, and Jonathan Rodriguez, Senior Member, IEEE

Abstract—Vehicular fog computing (VFC) has emerged as a promising solution to relieve the overload on the base station and reduce the processing delay during the peak time. The computation tasks can be offloaded from the base station to vehicular fog nodes by leveraging the under-utilized computation resources of nearby vehicles. However, the wide-area deployment of VFC still confronts several critical challenges, such as the lack of efficient incentive and task assignment mechanisms. In this paper, we address the above challenges and provide a solution to minimize the network delay from a contract-matching integration perspective. First, we propose an efficient incentive mechanism based on contract theoretical modeling. The contract is tailored for the unique characteristic of each vehicle type to maximize the expected utility of the base station. Next, we transform the task assignment problem into a two-sided matching problem between vehicles and user equipment. The formulated problem is solved by a pricing-based stable matching algorithm, which iteratively carries out the “propose” and “price-rising” procedures to derive a stable matching based on the dynamically updated preference lists. Finally, numerical results demonstrate that significant performance improvement can be achieved by the proposed scheme.

Index Terms—Vehicular fog computing, resource allocation, task assignment, contract theory, matching theory.

Manuscript received June 1, 2018; revised November 17, 2018; accepted January 11, 2019. Date of publication January 23, 2019; date of current version April 16, 2019. This work was supported in part by the National Natural Science Foundation of China under Grant 61601181, in part by the Fundamental Research Funds for the Central Universities under Grant 2017MS001, in part by the European Union’s Horizon 2020 Research and Innovation Program under Grant 815178, in part by the European Union’s Horizon 2020 Research and Innovation Program under the Marie Skłodowska-Curie Grant 824019, and in part by the Sichuan Science and Technology Program under Grant 2019YFH0033. The review of this paper was coordinated by the Guest Editors of the Special Section on Fog/Edge Computing for Autonomous and Connected Cars. (Corresponding author: Junhao Feng.)

Z. Zhou, P. Liu, and J. Feng are with the State Key Laboratory of Alternate Electrical Power System with Renewable Energy Sources, School of Electrical and Electronic Engineering, North China Electric Power University, Beijing 102206, China (e-mail: zhenyu_zhou@ncepu.edu.cn; pengju_liu@ncepu.edu.cn; junhao_feng@ncepu.edu.cn).

Y. Zhang is with the Department of Informatics, University of Oslo, Oslo, Norway and also with the Simula Metropolitan Center for Digital Engineering, 1325 Lysaker, Norway (e-mail: yanzhang@ieee.org).

S. Mumtaz is with the Instituto de Telecomunicações, Aveiro 3810-193, Portugal (e-mail: smumtaz@av.it.pt).

J. Rodriguez is with the Instituto de Telecomunicações, Aveiro 3810-193, Portugal, and also with the University of South Wales, Pontypridd CF37 1DL, U.K. (e-mail: jonathan.rodriguez@southwales.ac.uk).

Digital Object Identifier 10.1109/TVT.2019.2894851

I. INTRODUCTION

WITH the rapid advancement of information and communication technologies, there arises a critical issue that both the data rate and computation demands grow exponentially. For example, emerging 5G applications such as real-time video streaming, augmented reality, interactive gaming, and self driving, require advanced data communication, computation, and storage techniques to handle the complicated data processing and storage operations [1]. This poses a new challenge on the conventional cloud computing paradigm. It is difficult to guarantee the stringent quality of service (QoS) and quality of experience (QoE) requirements due to the long distance between user equipments (UEs) and remote data centers [2]. Edge computing which extends the computation capability to the close proximity of UEs has been proposed as a complementary solution [3], [4]. In [5], Cui *et al.* investigated the energy minimization problem in cache-assisted mobile edge computing (MEC), and proposed a joint caching and offloading mechanism. Guo *et al.* proposed an energy-efficient resource allocation scheme for multi-user MEC to optimally allocate the communication and computation resources [6]. However, in order to cover a large-scale geographic area, a massive number of high-cost energy-inefficient servers have to be deployed and maintained, which inevitably results in significant capital expenditure (CAPEX) and operational expenditure (OPEX). Furthermore, considering the dynamically time-varying demands, the dense deployment of servers will lead to huge resource wastage during the off-peak time. Therefore, how to accommodate the ever-increasing demand in communication and computation with moderate costs via a demand-adaptation approach remains an open problem.

An alternative choice is to exploit the under-utilized resources of nearby vehicles. Particularly, future vehicles will be equipped with more powerful onboard computers, larger-capacity data storage units, and more advanced communication modules for the sake of improving driving safety, convenience, and satisfaction [7], [8]. Hence, the tremendous computation resources provided by a large group of vehicles can be aggregated and utilized to alleviate network congestion during the peak time without deploying additional servers. For example, an array of parked vehicles can serve as fog node and provide real-time computation capability augmentation [9]. Moreover, the computation tasks of UEs can be directly offloaded to vehicles without going through the base station to further reduce the transmission

delay. This new computing paradigm is known as vehicular fog computing (VFC) [10], which is a beneficial complement to edge computing and cloud computing.

However, despite the above-mentioned advantages, the wide area deployment of VFC still confronts several critical challenges, which are summarized as follows.

First, there lacks an effective incentive mechanism for vehicles to serve as fog nodes. Most of previous studies have assumed that vehicles will share their computation resources unconditionally [11]. This assumption is too optimistic in practical implementation. Due to the cost incurred by task processing, self-interested vehicles are reluctant to serve as fog nodes unless they are well compensated. Furthermore, a vehicle's private information, such as the preference towards resource sharing and the total amount of available resources, is asymmetric, i.e., it is known by the vehicle itself but unavailable for the base station. This is called the scenario of information asymmetry. Therefore, it is of vital importance to develop an incentive mechanism, which can effectively optimize the economic benefit of the network operator or the base station under information asymmetry.

Second, there lacks a low-complexity near-optimal task assignment mechanism. With the existence of multiple vehicles and UEs, a critical challenge is how to assign the computation tasks of UEs to vehicles such that the total network delay can be minimized. Since UEs are owned by independent entities, it is highly possible that they have completely different interests and even prefer conflicting task assignment decisions. Therefore, a computation task can be implemented if and only if all the UEs have reached an agreement on the task assignment decision. Otherwise, some UEs can simply achieve a better performance by ignoring the decision. This is different from conventional VFC task assignment problems where the optimization is performed unilaterally [12], [13].

Accordingly, these challenges motivate us to develop a two-stage computation resource allocation and task assignment approach by combining contract theory and matching theory. In the first stage, in order to motivate vehicles to share their resources, the base station designs a contract, which specifies the relationship between the performance, i.e., the amount of computation resources required from a vehicle, and the reward, i.e., the payment to the vehicle for its contribution. In the contract, each distinct performance-reward association is defined as a contract item, and a contract generally contains a great variety of contract items. Then, the base station broadcasts the contract, and each vehicle chooses its desired contract item to maximize its payoff. In the second stage, the vehicles which have signed the contract with the base station serve as fog nodes. The task assignment problem is modeled as a two-sided matching game, in which the UEs rank the vehicles by considering transmission delay, task execution latency, task size, and matching cost. A stable matching between UEs and vehicles is derived by using the proposed pricing-based matching approach. To the best of the authors' knowledge, this is the first work which investigates the computation resource allocation and task assignment problem in VFC from a contract-matching integration perspective. The main contributions of this work are summarized as follows:

- *Contract-based incentive mechanism design:* We propose an efficient incentive mechanism based on

contract theoretical modeling. The contract is tailored for the unique characteristic of each vehicle type to maximize the expected utility of the base station under the constraints of individual rationality (IR), incentive compatibility (IC), and monotonicity. To make the problem tractable, the total number of IR and IC constraints are firstly reduced by exploring the relationships between adjacent vehicle types. Then, the simplified problem is solved by using Karush-Kuhn-Tucker (KKT) conditions. We also consider the scenario without information asymmetry and derive the corresponding optimal contract, which is used as a performance benchmark.

- *Matching-based computation task assignment:* The task assignment problem is intractable due to the combinatorial nature. To reduce the complexity, we transform the task assignment problem into a two-sided matching problem based on the problem structure, which involves a matching between vehicles on one side and UEs on the other side. Then, we propose a pricing-based stable matching algorithm to solve the task assignment problem, which iteratively carries out the *propose* and *price-rising* procedures to derive a stable matching based on the dynamically updated preference lists.
- *Theoretical analysis and performance validation:* We provide a comprehensive theoretical analysis on contract feasibility, matching convergence, matching stability, matching optimality, and computation complexity. The contract feasibility and efficiency as well as the network delay performance are evaluated by conducting a series of simulations under different scenarios. Numerical results demonstrate that the proposed algorithm can approach the optimal performance of the exhaustive searching algorithm, while the computation complexity is several orders of magnitude lower.

The remaining parts of the paper are summarized as follows. A comprehensive review of related works is provided in Section II. The overall system model is introduced in Section III. Section IV presents the contract-based incentive mechanism design. Section V elaborates the matching-based task assignment mechanism. Section VI provides the simulation results. The conclusion is given in Section VII.

II. RELATED WORKS

With the rapid proliferation of vehicles, the studies on VFC have received considerable attentions from both industry and academia. Hou *et al.* investigated the feasibility of VFC and provided a quantitative analysis among capacity, vehicle mobility, and connectivity [10]. Feng *et al.* proposed a novel framework named autonomous vehicular edge (AVE) to increase the computation capabilities of vehicles in a decentralized manner [14]. In [15], Satyanarayanan *et al.* explored how to build a shared real-time information system for vehicles to enable situational awareness based on the convergence of three technology trends. Xiao *et al.* investigated the feasibility of VFC, and developed a cost-effective on-demand VFC architecture by leveraging the mobility of vehicles [16]. In [17], Zhu *et al.* proposed a low-latency quality-enhanced task assignment solution

named fog following me (Folo) for VFC. The task assignment across stationary and mobile fog nodes is formulated as a joint optimization problem, and solved by exploiting mixed integer linear programming. As we can observe, these works rely on a common assumption that all the vehicles are willing to act as fog nodes, and the incentive issues have been neglected.

There exist some works which have already investigated the incentive design problem in cloud/edge/fog computing. In [18], Luong *et al.* proposed a comprehensive literature survey of pricing-based incentive mechanisms for resource allocation in cloud-enabled wireless networks. Liu *et al.* considered the computation offloading problem in MEC, and provided a Stackelberg game-based pricing scheme to stimulate edge server owners while optimizing the utility of the cloud service operator [19]. In [20], Su *et al.* developed a Stackelberg game-based pricing scheme to coordinate the competition and cooperation among moving vehicles, parked vehicles, and road side unit (RSU). In the Stackelberg game, the leaders, i.e., the RSU and parking area, have the perfect knowledge of the moving vehicles and content delivery costs. In addition, they also know each side's optimal strategy. However, most of current works rely on symmetric information, and are not applicable to the scenario of information asymmetry.

Contract theory is regarded as a powerful tool from microeconomics to cope with information asymmetry. A vehicle's private information can be effectively elicited because the contract item is incentive compatible, i.e., the payoff of a vehicle is maximized if and only if it selects the contract item designed for its type. Contract theory has already been widely applied in the optimization of wireless networks. In [21], Du *et al.* proposed a contract-based user association approach for traffic offloading in software-defined heterogeneous networks. Xu *et al.* developed an energy-efficient relay selection scheme by exploiting contract theoretical modeling [22]. Other application scenarios include cognitive radios [23], mobile crowdsourcing [24], and small-cell caching systems [25]. Several previous works have already compared Stackelberg game with contract theory. In [26], Duan *et al.* investigated the incentive mechanism design for smartphone collaboration. The Stackelberg game is used to model the cooperation game for data acquisition, where the shared tasks and the corresponding rewards for collaborators are homogeneous. In comparison, the contract theory is used to motivate cooperation in distributed computing, where computation efficiency and task amount for collaborators heterogeneous. In [24], Liu *et al.* demonstrated that the contract theory provides better profit for the base station than the Stackelberg game due to the fact that the contract is completely designed by the base station, which acts as a monopolist in the market. In the Stackelberg game, since the information is symmetric, the followers also know the action of the leader, and can optimize their payoffs accordingly.

Another critical challenge in VFC is how to assign the computation tasks to vehicular fog nodes. Numerous studies have addressed the task assignment problem with different optimization approaches, e.g., matching theory [1], [27], coalitional game [28], Stackelberg game [29], and multi-player noncooperative game [30]. Compared to other solutions, matching theory is more suitable to handle the heterogeneous preferences of UEs. Specifically, the task assignment problem can be modeled as a

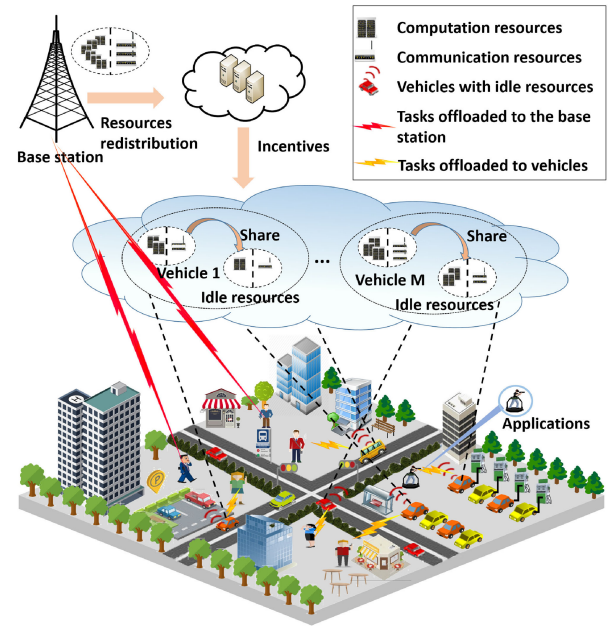


Fig. 1. The VFC framework.

two-sided matching game between UEs and vehicles, and solved in a self-organizing and self-optimizing fashion. Matching theory has already been employed to address various combinatorial problems with mutual preferences in Internet of things (IoT) fog networks [31], device-to-device networks [12], and vehicular content distribution networks [1], etc.

Based on the above literature review, we can conclude that there lacks a uniform framework to address the resource allocation and task assignment optimization problem in VFC from a contract-matching integration perspective. Specifically, how to combine these two powerful tools to minimize the overall network delay requires further investigation.

III. SYSTEM MODEL

The VFC framework is shown in Fig. 1. In each cell, there exists a base station which takes charge of intra-cell communication resource coordination, computation resource allocation and task assignment. During the peak time when the base station is overwhelmed by the incoming computation demands, a group of vehicles are employed to act as fog nodes and relieve the overload problem via task offloading. Any vehicle with idle computation resources is able to act as a fog node by sharing its resources for task processing. With a properly-designed incentive mechanism, each vehicle can actively adjust the amount of resources to be shared in order to maximize its individual payoff. The details for how to design the incentive mechanism will be illustrated in Section IV.

In the same cell, there also exist numerous UEs. Each UE generates a series of computation tasks, each of which can be either processed by the base station or offloaded to a vehicular fog node. The details for how to model the interactions between UEs and vehicles, and how to derive a low-complexity sub-optimal task assignment solution will be illustrated in Section V.

For the sake of simplicity, we adopt a time-slot model in which the time is slotted into discrete intervals [22]. The

optimization process is carried out in a slot-by-slot fashion. The set of vehicles and the set of UEs within the coverage of the base station remain fixed within each slot, and vary across different slots, which are denoted as $\mathcal{V}_M = \{V_1, \dots, V_m, \dots, V_M\}$ and $\mathcal{U}_N = \{U_1, \dots, U_n, \dots, U_N\}$, respectively. During each slot, it is assumed that each UE, e.g., U_n , has a computation task to be processed. The key attributes of the task can be described by a triplet $\{D_n, C_n, \tau_n\}$, where D_n represents the task data size, C_n is the required computation resource, i.e., the computation size, and τ_n represents the delay constraint.

Remark 1: The system model considered in this work can also be extended to the scenario that a UE has multiple computation tasks to be processed per slot. In this case, the multiple tasks can be aggregated and considered as a single task with larger data size and higher computation demand. If UE U_n has no task, we can simply set $D_n = C_n = \tau_n = 0$.

Remark 2: Our model is different from the conventional MEC model where UEs offload their tasks to an edge server. First, the location of the edge server is fixed, while both the fog nodes and UEs considered in our model can be mobile. Second, the incentive issues have been largely neglected in MEC since both the communication and computation infrastructures are deployed and owned by the same network operator. However, in VFC where vehicles are owned by individuals, the incentive issues must be taken into consideration. Third, in the MEC model, UEs within the same cell can only connect to one or at most two base stations. In comparison, due to the high density of vehicles, a UE may be surrounded by multiple vehicles, and the candidate vehicles of different UEs are overlapped with each other. Last but not least, the MEC model is more suitable for the centralized task assignment scenario where the tasks of all the UEs within the same cell are assigned to the same edge server. Our model emphasizes on the decentralized task assignment scenario where the tasks of UEs are assigned to a group of distributed vehicles.

IV. CONTRACT-BASED INCENTIVE MECHANISM DESIGN

In this section, we propose a contract-based incentive mechanism to motivate vehicles to share their computation resources for task offloading. First, we introduce the vehicle type model. Second, the utility functions of the base station and vehicles are introduced, and the resource allocation problem is formulated. Third, we elaborate how to derive the optimal contract under information asymmetry. Finally, the optimal contract design without information asymmetry is provided.

A. Vehicle Type Modeling

The preference of a vehicle towards resource sharing is quantified as its vehicle type. A vehicle with a higher type is more willing to share its resources and serve as a fog node compared to a vehicle with a lower type. Thus, it is intuitive for the base station to employ higher-type vehicles. Since the number of vehicles in a cell is usually finite, the set of vehicle types belongs to a discrete and finite space. The vehicle type is defined as follows:

Definition 1. (Vehicle Type): The M vehicles in set $\mathcal{V}_M$ can be sorted in an ascending order based on their preferences and classified into K types. Denote the set of vehicle types as

$\mathcal{K} = \{1, \dots, k, \dots, K\}$, and denote the set of corresponding resource sharing capability as $\Theta = \{\theta_1, \dots, \theta_k, \dots, \theta_K\}$, which is given by

$$\theta_1 < \dots < \theta_k < \dots < \theta_K, \quad k \in \mathcal{K} \quad (1)$$

Then, we show how to derive the explicit expression of the vehicle type. For vehicle V_m , denote C_m as the computation size of the local task to be processed. Due to resource sharing, the processing delay will be increased, which is given by

$$\Delta\tau_m = \frac{C_m}{\delta_{m,0} - \delta_m} - \frac{C_m}{\delta_{m,0}} \leq \Delta\tau_{m,max}, \quad (2)$$

where $\delta_{m,0}$ and δ_m represent the total available computation resource and the shared resource, respectively. The inequality specifies that the increased delay should be less than or equal to a threshold $\Delta\tau_{m,max}$ to satisfy QoS or QoE requirements.

Through some manipulations of (2), we can derive the upper bound of δ_m as

$$\delta_m \leq \frac{\delta_{m,0}^2 \Delta\tau_{m,max}}{\delta_{m,0} \Delta\tau_{m,max} + C_m} = \delta_m^{upper}, \quad (3)$$

where δ_m^{upper} is the maximum amount of resources that can be shared. We assume that δ_m^{upper} falls into a continuous closed interval $[\delta_{min}, \delta_{max}]$, where δ_{min} and δ_{max} represent the minimum and maximum values of δ_m^{upper} , $\forall m \in \mathcal{M}$, respectively. Then, the interval is divided into K subintervals with the same length, and the lower bound of the k -th subinterval is defined as θ_k , which is given by

$$\theta_k = \delta_{min} + \frac{k-1}{K}(\delta_{max} - \delta_{min}). \quad (4)$$

The type of vehicle V_m is said to be k if $\theta_k \leq \delta_m^{upper} < \theta_{k+1}$.

Remark 3: From (4), we can infer that θ_k increases with $\delta_{m,0}$ and $\Delta\tau_{m,max}$, and decreases with C_m . This definition is consistent with practical situations. For example, a vehicle with light local tasks and abundant idle resources can share more resources. As a result, it can gain a higher profit and thus has a higher preference towards resource sharing.

In the scenario of information asymmetry, the base station does not know the precise information of each vehicle's type. Instead, only the statistical knowledge of the vehicle type is available via long term measurements or historical observations. We assume that the base station only knows that there are a total of K types of vehicles and each vehicle $V_m \in \mathcal{V}_M$ belongs to type k with the same probability λ_k , i.e., $\sum_{k=1}^K \lambda_k = 1$.

B. Contract Formulation

Instead of offering the same contract item to vehicles with different types, the base station can design up to K contract items for K vehicle types, i.e., one contract item per type. For instance, the contract item dedicated for type k vehicle is denoted as (δ_k, π_k) , where δ_k denotes the required computation resources, and π_k is the corresponding reward. The whole contract is denoted as $\mathcal{C} = \{(\delta_k, \pi_k), \forall k \in \mathcal{K}\}$.

Assuming the total amount of computation tasks that can be processed by the base station during a time interval T is C_{BS} , we have $C_{BS} = \delta_{BS}T$. Here, δ_{BS} is the computation capability of the base station per second. We assume that the benefit of the

base station is positively related to the reduced delay, which is valid since the motivation of exploiting VFC is to reduce the computation delay. For the purpose of simplicity, the linear function is utilized, which has also been employed in numerous works [32], [33]. Nevertheless, the proposed scheme can also be extended to more complicated scenarios with nonlinear functions. By signing the contract item (δ_k, π_k) with type k vehicle, the benefit of the base station is given by

$$R_{BS}(\delta_k) = r_{BS} \left(\frac{C_{BS}}{\delta_{BS}} - \frac{C_{BS}}{\delta_k + \delta_{BS}} \right) = r_{BS} T \frac{\delta_k}{\delta_{BS} + \delta_k}, \quad (5)$$

where r_{BS} is the unit benefit brought by the reduced delay. It is noted that although (5) is defined as a linear function of the reduced delay, it is actually a concave function of the variable δ_k , which is nonlinear.

With K types and M vehicles, the expected utility of the base station is calculated as

$$U_{BS}(\{\delta_k\}, \{\pi_k\}) = M \sum_{k=1}^K \lambda_k [R_{BS}(\delta_k) - \pi_k]. \quad (6)$$

Remark 4: A contract item $(\delta_k = 0, \pi_k = 0)$ means that type k vehicle has no desire to share its resources. Moreover, the contract item must also guarantee that the utility of the base station is nonnegative, i.e., $R_{BS}(\delta_k) - \pi_k \geq 0$. Otherwise, the base station has no incentive to sign the contract with type k vehicle.

The utility function of type k vehicle which accepts the contract item (δ_k, π_k) is given by

$$U_k^V(\delta_k, \pi_k) = \theta_k \pi_k - \delta_k, \quad (7)$$

where θ_k characterize the weight of π_k to type k vehicle. A higher-type vehicle has a larger weight due to its higher preference towards resource sharing.

The expected social welfare is the total sum utility of the base station and the M vehicles, which is given by

$$U_s(\{\delta_k\}, \{\pi_k\}) = U_{BS}(\{\delta_k\}, \{\pi_k\}) + M \sum_{k=1}^K \lambda_k U_k^V(\delta_k, \pi_k). \quad (8)$$

The objective of the base station is to maximize its utility under the scenario of asymmetric information via the optimization of each contract item. The corresponding optimization problem is formulated as

$$\begin{aligned} \mathbf{P1} : \quad & \max_{(\{\delta_k\}, \{\pi_k\})} U_{BS}(\{\delta_k\}, \{\pi_k\}) \\ \text{s.t.} \quad & C_1 : \theta_k \pi_k - \delta_k \geq 0, \forall k \in \mathcal{K}, (IR) \\ & C_2 : \theta_k \pi_k - \delta_k > \theta_{k'} \pi_{k'} - \delta_{k'}, \\ & \quad \forall k, k' \in \mathcal{K}, k \neq k', (IC) \\ & C_3 : 0 \leq \delta_1 < \dots < \delta_k < \dots < \delta_K, \forall k \in \mathcal{K}, \\ & C_4 : \delta_k \leq \theta_k, \forall k \in \mathcal{K}, \end{aligned} \quad (9)$$

where C_1 , C_2 , and C_3 represent the IR, IC, and monotonicity constraints, respectively. C_4 represents the upper bound of δ_k .

Definition 2: The IR, IC, and monotonicity constraints are defined as follows:

- *Individual rationality (IR) constraint:* Type k vehicle, $\forall k \in \mathcal{K}$, will get a nonnegative payoff if it selects the contract item (δ_k, π_k) .
- *Incentive compatibility (IC) constraint:* The IC constraint ensures the self-revealing property of the contract. For instance, type k vehicle, $\forall k \in \mathcal{K}$, will get the maximum payoff if and only if it selects the contract item (δ_k, π_k) designed for its own type.
- *Monotonicity constraint:* The reward of type k vehicle, $\forall k \in \mathcal{K}$, should be higher than that of type $k-1$ vehicle, and lower than that of type $k+1$ vehicle.

Based on the IR, IC, and monotonicity constraints, we have *Lemma 1:* For any $k, k' \in \mathcal{K}$, if $\theta_k > \theta_{k'}$, then $\delta_k > \delta_{k'}$ and $\pi_k > \pi_{k'}$. $\pi_k = \pi_{k'}$ and $\delta_k = \delta_{k'}$ if and only if $\theta_k = \theta_{k'}$.

Lemma 2: For any $(\delta_k, \pi_k) \in \mathcal{C}$, the following inequalities hold

$$\begin{aligned} 0 &\leq \pi_1 \leq \dots \leq \pi_k \leq \dots \leq \pi_K, \\ 0 &\leq \delta_1 \leq \dots \leq \delta_k \leq \dots \leq \delta_K, \\ 0 &\leq U_1^V \leq \dots \leq U_k^V \leq \dots \leq U_K^V. \end{aligned} \quad (10)$$

Proof: A similar proof of Lemma 1 and Lemma 2 can be found in [34]. The details are omitted here due to space limitation. ■

Based on Lemma 1 and Lemma 2, we define the sufficient and necessary conditions for contract feasibility.

Theorem 1: Contract feasibility: The contract $\mathcal{C} = \{(\delta_k, \pi_k), \forall k \in \mathcal{K}\}$ is feasible if and only if all the following conditions are satisfied:

- $0 \leq \pi_1 \leq \dots \leq \pi_k \leq \dots \leq \pi_K$ and $0 \leq \delta_1 \leq \dots \leq \delta_k \leq \dots \leq \delta_K$;
- $\theta_1 \pi_1 - \delta_1 \geq 0$;
- For any $k \in \{2, \dots, K\}$, $\delta_{k-1} + \theta_{k-1}(\pi_k - \pi_{k-1}) \leq \delta_k \leq \delta_{k-1} + \theta_k(\pi_k - \pi_{k-1})$.

Proof: The detailed proof of Theorem 1 is omitted here due to space limitation. A similar proof can be found in Appendix D of [23]. ■

C. Optimal Contract Design Under Information Asymmetry

Problem **P1** involves K IR constraints and $K(K-1)$ IC constraints. To provide a tractable solution, the following procedures are carried out to simplify the problem.

Step 1: IR Constraints Elimination

For type k vehicle, $k \in \mathcal{K}, k \neq 1$, we can derive

$$U_k^V \geq U_{k-1}^V \geq U_1^V \geq 0, \quad (11)$$

where the first inequality is due to the IC constraint, the second inequality is based on Lemma 2, and the third inequality is due to the IR constraint. Hence, the IR constraint of type k vehicle holds automatically as long as the IR constraint of type 1 vehicle is guaranteed.

Step 2: IC Constraints Elimination

We define the IC constraints between type k and type k' , $k' \in \{1, \dots, k-1\}$, as downward incentive constraints (DICs). Similarly, the IC constraints between type k and type k'' , $k'' \in$

$\{k+1, \dots, K\}$, are defined as upward incentive constraints (UICs). In the following, we show that both the DICs and UICs can be reduced.

We consider three adjacent vehicle types, i.e., $\theta_{k-1} < \theta_k < \theta_{k+1}$, which satisfy

$$\theta_{k+1}\pi_{k+1} - \delta_{k+1} \geq \theta_{k+1}\pi_k - \delta_k, \quad (12)$$

$$\theta_k\pi_k - \delta_k \geq \theta_k\pi_{k-1} - \delta_{k-1}, \quad (13)$$

where (12) denotes the DIC between type $k+1$ and k , and (13) denotes the DIC between type k and $k-1$.

By combining $\pi_{k+1} \geq \pi_k \geq \pi_{k-1}$, we have

$$\theta_{k+1}\pi_{k+1} - \delta_{k+1} \geq \theta_{k+1}\pi_{k-1} - \delta_{k-1}. \quad (14)$$

Therefore, if the DIC between type $k+1$ and k holds, then the DIC between type $k+1$ and $k-1$ also holds. The DIC constraints can be extended downward from type $k-1$ to type 1, which are given by

$$\begin{aligned} \theta_{k+1}\pi_{k+1} - \delta_{k+1} &\geq \theta_{k+1}\pi_{k-1} - \delta_{k-1} \\ &\geq \dots \\ &\geq \theta_{k+1}\pi_1 - \delta_1. \end{aligned} \quad (15)$$

Thus, we demonstrate that if the DICs between adjacent types hold, then all the DICs hold automatically. Similarly, we can demonstrate that if the UICs between adjacent types hold, then all the UICs hold automatically.

Based on the above analysis, the K IR constraints and $K(K-1)$ IC constraints can be reduced to 1 and $K-1$, respectively. Furthermore, we have the following properties:

Proposition 1: In order to maximize the utility of the base station, the optimal contract item for type 1 vehicle, i.e., (δ_1^*, π_1^*) , must enforce

$$U_1^V(\delta_1^*, \pi_1^*) = \theta_1\pi_1^* - \delta_1^* = 0. \quad (16)$$

Proof: The proof is based on reduction to absurdity. Assuming $\theta_1\pi_1^* - \delta_1^* > 0$, then the base station can improve its own utility by either decreasing π_1^* or increasing δ_1^* until $\theta_1\pi_1 - \delta_1 = 0$ while simultaneously satisfying the conditions of contract feasibility. Then, we have

$$R_{BS}(\hat{\delta}_1) - \hat{\pi}_1 > R_{BS}(\delta_1^*) - \pi_1^*, \quad (17)$$

which contradicts with the assumption that (δ_1^*, π_1^*) is the optimal contract item. Hence, we must have $\theta_1\pi_1^* - \delta_1^* = 0$. This completes the proof of Proposition 1. ■

Proposition 2: The optimal contract item for any type k vehicle (δ_k^*, π_k^*) , $k = 2, \dots, K$, satisfies the following equality condition:

$$\delta_k^* = \delta_{k-1}^* + \theta_k(\pi_k^* - \pi_{k-1}^*). \quad (18)$$

Proof: From the IC constraint, we have

$$\delta_k^* \leq \delta_{k-1}^* + \theta_k(\pi_k^* - \pi_{k-1}^*), \quad k = 2, \dots, K \quad (19)$$

Then, the base station can further improve its own utility by either decreasing π_k^* or increasing δ_k^* until the equality holds, which does not violate the conditions of contract feasibility. This completes the proof of Proposition 2. ■

Thus, based on constraint elimination, Proposition 1 and Proposition 2, **P1** can be rewritten as

$$\begin{aligned} \mathbf{P2}: \quad & \max_{\{\delta_k\}, \{\pi_k\}} U_{BS}(\{\delta_k\}, \{\pi_k\}), \\ \text{s.t.} \quad & C_1: \theta_1\pi_1 - \delta_1 = 0, (IR) \\ & C_2: \delta_k = \delta_{k-1} + \theta_k(\pi_k - \pi_{k-1}), 2 \leq k \leq K, (IC) \\ & C_3, C_4, \forall k \in \mathcal{K}. \end{aligned} \quad (20)$$

We can easily prove that **P2** is a convex programming problem by checking the Hessian matrix. Thus, **P2** can be solved by applying KKT conditions. The Lagrangian associated with **P2** is given by

$$\begin{aligned} \mathcal{L}(\{\delta_k\}, \{\pi_k\}, \{\mu_k\}, \{\rho_k\}, \{\beta_k\}) \\ = U_{BS}(\{\delta_k\}, \{\pi_k\}) + \mu_1(\theta_1\pi_1 - \delta_1) \\ + \sum_{k=2}^K \mu_k \left(\theta_k(\pi_k - \pi_{k-1}) + \delta_{k-1} - \delta_k \right) \\ + \rho_1\delta_1 + \sum_{k=2}^K \rho_k(\delta_k - \delta_{k-1}) + \sum_{k=1}^K \beta_k(\delta_k - \theta_k), \end{aligned} \quad (21)$$

where μ_1 is the Lagrange multiplier corresponding to constraint C_1 , $\{\mu_k, k = 2, \dots, K\}$, $\{\rho_k, \forall k \in \mathcal{K}\}$, and $\{\beta_k, \forall k \in \mathcal{K}\}$ are the vectors of Lagrange multipliers corresponding to constraints C_2 , C_3 , and C_4 , respectively. KKT conditions are summarized as follows:

- Primal constraints: $0 \leq \delta_1^*$; $\delta_{k-1}^* \leq \delta_k^*$, $\delta_k^* = \delta_{k-1}^* + \theta_k(\pi_k^* - \pi_{k-1}^*)$, $\forall k \in \mathcal{K}, k \neq 1$; $\delta_1^* = \theta_1\pi_1^*$; $\delta_k^* \leq \theta_k$, $\forall k \in \mathcal{K}$;
- Dual constraints: $\mu_k^* \geq 0$, $\rho_k^* \geq 0$ and $\beta_k^* \geq 0$, $\forall k \in \mathcal{K}$;
- Complementary slackness: $\rho_1^*\delta_1^* = 0$; $\rho_k^*(\delta_k^* - \delta_{k-1}^*) = 0$, $\forall k \in \mathcal{K}, k \neq 1$; $\beta_k^*(\delta_k^* - \theta_k) = 0$, $\forall k \in \mathcal{K}$;
- The first-order conditions of the Lagrangian is

$$\left\{ \begin{aligned} \frac{\partial \mathcal{L}}{\partial \delta_k} &= \frac{\partial R_{BS}(\delta_k)}{\partial \delta_k} - \mu_k + \mu_{k+1} + \rho_k - \rho_{k+1} \\ &\quad + \beta_k = 0, \forall k \in \mathcal{K}, k \neq K, \\ \frac{\partial \mathcal{L}}{\partial \delta_K} &= \frac{\partial R_{BS}(\delta_K)}{\partial \delta_K} - \mu_K + \rho_K + \beta_K = 0, \\ \frac{\partial \mathcal{L}}{\partial \pi_k} &= -\lambda_k + \mu_k\theta_k - \mu_{k+1}\theta_{k+1} = 0, \\ &\quad \forall k \in \mathcal{K}, k \neq K, \\ \frac{\partial \mathcal{L}}{\partial \pi_K} &= -\lambda_K + \mu_K\theta_K = 0. \end{aligned} \right. \quad (22)$$

The contract design and optimization is handled by the centralized base station, and the detailed process is summarized as part of the Algorithm 1.

D. Optimal Contract Design Without Information Asymmetry

If there exists a selfish base station which is perfectly aware of every vehicle's type, it can further increase its profit by exploiting the complete information. The contract has to ensure that the payoff of each vehicle is non-negative. Otherwise, the vehicles have no incentive to accept the contract items. To this end, the

contract item also has to meet the IR constraint. Furthermore, the contract item has to satisfy the following property:

Theorem 2: In the contract design without information asymmetry, any optimal contract item $(\delta_k^*, \pi_k^*) \in \mathcal{C}$ should satisfy $\theta_k \pi_k^* = \delta_k^*$. That is, the payoff for any vehicle is zero.

Proof: Theorem 2 can be proved by contradiction. Given an optimal contract item (δ_k^*, π_k^*) , if $\theta_k \pi_k^* - \delta_k^* > 0$, then the base station can increase its utility by decreasing π_k^* to $\hat{\pi}_k$ which satisfies $\theta_k \hat{\pi}_k - \delta_k^* = 0$. This contradicts with the assumption that (δ_k^*, π_k^*) is optimal. ■

Thus, by enforcing the utility of each vehicle to be zero, **P2** can be written as

$$\begin{aligned} \mathbf{P3} : \quad & \max_{(\{\delta_k\}, \{\pi_k\})} U_{BS}(\{\delta_k\}, \{\pi_k\}), \\ \text{s.t.} \quad & \tilde{C}_1 : \theta_k \pi_k - \delta_k = 0, \forall k \in \mathcal{K}, \\ & C_3, C_4. \end{aligned} \quad (23)$$

P3 can also be solved by using KKT conditions.

Theorem 3: In the contract design without information asymmetry, for any type k vehicle, $k \in \mathcal{K}$, the optimal reward is $\pi_k^* = 1$ regardless of θ_k .

Proof: From Theorem 2, we have $\theta_k \pi_k^* - \delta_k^* = 0$. Furthermore, from the upper bound of $\delta_k \leq \theta_k$, we can derive $\delta_k^* = \theta_k$ since the base station can always increase δ_k to increase its utility until the equality holds. Thus, we have $\theta_k \pi_k^* - \theta_k = 0$, and $\pi_k^* = 1$. ■

V. MATCHING-BASED TASK ASSIGNMENT

In this section, we first introduce the task assignment model and the problem formulation. Then, we introduce the proposed pricing-based matching algorithm. Next, we provide a comprehensive theoretical analysis on convergence, stability, optimality, and complexity.

A. Task Assignment Model

1) *Task Transmission Delay:* After the first-stage resource sharing, the UEs can offload their computation tasks to vehicles which serve as fog nodes. In the offloading mode, data can be directly transmitted from UEs to vehicles to reduce the total number of transmission hops. We assume that each UE is allocated with an orthogonal spectrum resource block such that the co-channel interference among UEs can be ignored. Furthermore, the large-scale fading and the small-scale fading are modeled by using the Rayleigh fading model and the free-space propagation path-loss model, respectively. If the task of UE U_n is offloaded to vehicle V_m , the signal to noise ratio (SNR) of the received signal at vehicle V_m is given by

$$\gamma_{m,n} = \frac{p_n d_{n,m}^{-\alpha} h_{n,m}^2}{N_0}, \quad (24)$$

where p_n denotes the transmission power of UE U_n . $d_{n,m}$ is the transmission distance between UE U_n and vehicle V_m . α is the path-loss exponent. $h_{n,m}$ represents the Rayleigh channel coefficient with a complex Gaussian distribution. N_0 denotes the power noise.

Hence, the transmission time required by UE U_n for uploading its task with size D_n can be obtained as

$$T_{n,m}^t = \frac{D_n}{B_{n,m} \log_2(1 + \gamma_{n,m})}, \quad (25)$$

where $B_{n,m}$ refers to the bandwidth of the link between UE U_n and vehicle V_m .

We assume that the vehicles travel on a two-lane two-directional road. Due to the fast vehicle mobility, vehicle V_m might move out of the communication range of UE U_n during data transmission, which results in an offloading failure. Denote the dwell time of V_m inside the communication range of U_n as $\tau_{n,m}^o$. An offloading failure occurs if $\tau_{n,m}^o < T_{n,m}^t$. Therefore, $\tau_{n,m}^o$ also represents the delay constraint of data transmission because U_n can only transmit data to V_m when they remain connected. That is, an offloading request is admissible if and only if $T_{n,m}^t < \tau_{n,m}^o$.

To estimate the vehicle dwell time, a simple way is to use the average velocity. Assuming that UE U_n is located along the road side and its communication range is a circle with a diameter d_n , $\tau_{n,m}^o$ can be calculated as

$$\tau_{n,m}^o = \hat{d}_{n,m} / \bar{v}_m, \quad (26)$$

where $\hat{d}_{n,m}$ denotes the distance between the location of V_m and the endpoint of the circle's diameter in the vehicle heading direction, and $\bar{v}_m$ denotes the average velocity of V_m .

Remark 5: Both $\hat{d}_{n,m}$ and $\bar{v}_m$ can be estimated from the GPS data [35]. For example, if V_m moves in the centrifugal direction to leave the communication area of V_n , $\hat{d}_{n,m}$ is calculated as $\hat{d}_{n,m} = \frac{1}{2d_n} - d_{n,m}$. Otherwise, if V_m moves in the centripetal direction, we have $\hat{d}_{n,m} = \frac{1}{2d_n} + d_{n,m}$. We have not put any restriction on the mobility models of vehicles. As long as the vehicle dwell time can be obtained, the proposed matching-based task assignment scheme can be adaptable for different mobility models.

Remark 6: In this work, we assume that the GPS information is known to the base station. This assumption has also been accepted and employed in a number of previous works [8], [14]. Furthermore, even if the GPS information is unavailable to the base station, vehicle positions can still be obtained based on mobility prediction techniques [36]–[39].

2) *Task Execution Delay:* If the type of vehicle V_m is k , i.e., the amount of computation resource is $\delta_m = \delta_k^*$, the execution time required to process the task of UE U_n is calculated as

$$T_{n,m}^c = \frac{C_n}{\delta_m} = \frac{C_n}{\delta_k^*}. \quad (27)$$

The transmission latency from V_m to U_n is ignored, due to the fact that the size of computation results is usually negligible compared to that of the input data, e.g., face recognition. If V_m has already moved out of the communication range of U_n , the results have to be forwarded firstly from V_m to the base station, and then sent from the base station to U_n .

The total delay when the task of UE U_n is assigned to vehicle V_m is given by

$$T_{n,m}^{\text{total}} = T_{n,m}^t + T_{n,m}^c. \quad (28)$$

3) *Problem Formulation*: The purpose of this work is to relieve the heavy burden of the base station and reduce the total network delay by leveraging the under-utilized computation resources of vehicles. Hence, we model the objective function as the total delay of the overall network, i.e., the sum delay of all the UEs. We investigate how to assign the tasks of UEs to vehicles such that the objective function is minimized. The task assignment decision is defined as follows:

Definition 3. (Task Assignment Decision): The task assignment decision between N UEs and M vehicles is defined as a $N \times M$ matrix $\mathbf{X}$, where its (n, m) -th element, i.e., $x_{n,m}$, is defined as a binary value. $x_{n,m} = 1$ means that the computation task of UE U_n is assigned to vehicle V_m . Otherwise, $x_{n,m} = 0$.

The problem is formulated as follows:

$$\begin{aligned} \text{P4: } \min_{\mathbf{X}} \quad & \sum_{n=1}^N \sum_{m=1}^M x_{n,m} T_{n,m}^{\text{total}} \\ \text{s.t. } \quad & C_5: \sum_{V_m \in \mathcal{V}_M} x_{n,m} \leq 1, \forall U_n \in \mathcal{U}_N, \\ & C_6: \sum_{U_n \in \mathcal{U}_N} x_{n,m} \leq 1, \forall V_m \in \mathcal{V}_M, \\ & C_7: T_{n,m}^{\text{total}} \leq \tau_n, \forall U_n \in \mathcal{U}_N, \forall V_m \in \mathcal{V}_M, \\ & C_8: T_{n,m}^t \leq \tau_{n,m}^o, \forall U_n \in \mathcal{U}_N, \forall V_m \in \mathcal{V}_M. \end{aligned} \quad (29)$$

Here, C_5 and C_6 guarantee that there is a one-to-one correspondence among UEs and vehicles. C_7 and C_8 denote the delay constraints of task assignment and task transmission, respectively.

To provide a tractable solution, the combinatorial problem **P4** can be transformed into a two-sided matching problem based on the problem structure. The matching problem can be represented as a triple $(\mathcal{U}_N, \mathcal{V}_M, \mathcal{F})$, where $\mathcal{U}_N$ and $\mathcal{V}_M$ are two distinct and finite sets of matching participants, i.e., UEs and vehicles, respectively. $\mathcal{F}$ represents the set of matching preferences of UEs. Particularly, each UE aims at minimizing its individual delay under the specified constraints. A one-to-one matching ϕ is defined as [40]:

Definition 4. (Matching): For the formulated matching problem $(\mathcal{U}_N, \mathcal{V}_M, \mathcal{F})$, the matching ϕ is a one-to-one correspondence from set $\mathcal{U}_N \cup \mathcal{M}_K$ onto itself under preference $\mathcal{F}$, i.e., $\phi(U_n) \in \mathcal{V}_M \cup \{U_n\}, \forall U_n \in \mathcal{U}_N$. $\phi(U_n) = V_m$ represents that UE U_n is matched with vehicle V_m , which is equivalent to $x_{n,m} = 1$. $\phi(U_n) = U_n$ means that U_n has not been matched with any vehicle, and the task of U_n will be handled by the base station.

B. Preference List Construction

In order to implement the two-sided matching, every UE has to construct its preference list by ranking vehicles from the other side in accordance with the preferences. For UE U_n , it can achieve different delay performances when being paired with different vehicles. Therefore, in order to minimize the total delay, we can define that the preference is inversely proportional to the total delay, e.g., $1/T_{n,m}^{\text{total}}$. The preference of U_n towards

V_m is calculated as

$$G_{n,m} |_{\phi(U_n)=V_m} = \frac{1}{T_{n,m}^{\text{total}}} - P_m, \quad (30)$$

where P_m is the price for utilizing the computation resource of V_m , the initial value of which is zero. The role of P_m is to resolve the matching conflict, which will be explained later. Here, $1/T^{\text{total}}$ is just used as an example. The preference model can be extended to more complicated expressions.

We introduce a complete, reflexive, and transitive binary preference relation [40], i.e., “ $\succ$ ”, to compare the preferences towards different vehicles. For instance, $V_m \succ_{U_n} V_{m'}$ represents that U_n prefers V_m to $V_{m'}$, which is given by

$$V_m \succ_{U_n} V_{m'} \Leftrightarrow G_{n,m} |_{\phi(U_n)=V_m} > G_{n,m'} |_{\phi(U_n)=V_{m'}}. \quad (31)$$

Furthermore, $V_m \succeq_{U_n} V_{m'}$ represents that U_n prefers V_m at least as well as $V_{m'}$, which is given by

$$V_m \succeq_{U_n} V_{m'} \Leftrightarrow G_{n,m} |_{\phi(U_n)=V_m} \geq G_{n,m'} |_{\phi(U_n)=V_{m'}}. \quad (32)$$

To obtain the entire preference list of U_n , we temporarily pair it with every vehicle in order to derive the preference for each combination. We use F_n to denote the preference list of U_n towards all the vehicles. F_n is obtained by sorting all the M vehicles in a descending order according to $G_{n,m} |_{\phi(U_n)=V_m}, \forall V_m \in \mathcal{V}_M$. The total set $\mathcal{F}$ is constructed as $\mathcal{F} = \{F_n, \forall U_n \in \mathcal{U}_N\}$.

C. Pricing-Based Stable Matching

After obtaining the preference for all the UEs in $\mathcal{U}_N$, the second-stage task assignment problem can be solved by using a pricing-based stable matching algorithm. The main parts of it are the *propose* and the *price rising* rules, which are defined as follows:

Definition 5. (Propose Rule): For any UE $U_n \in \mathcal{U}_N$, it proposes to its most preferred vehicle which ranks as the first place in its preference list F_n , e.g., V_m . We have $V_m \succ_{U_n} V_{m'}, \forall V_{m'} \in F_n, V_{m'} \neq V_m$.

Definition 6. (Price Rising Rule): For any vehicle $V_m \in \mathcal{V}_M$ that has received more than one proposal from UEs, it can raise its price to increase the matching cost, which is given by

$$P_m = P_m + \Delta P_m. \quad (33)$$

The matching process is implemented in an iterative fashion, and the detailed process is summarized as part of the Algorithm 1. It is noted that the matching-based task assignment can also be managed by the base station. Specifically, each UE uploads its preference list to the base station, and then a stable matching between UEs and vehicles is derived by the base station based on the preference lists of UEs. Eventually, the base station broadcasts the matching result to UEs, which offload their tasks to the corresponding vehicles accordingly. The implementation procedure is explained as follows.

Phase 1: Matching Preference Initialization

- Calculate F_n for each UE $U_n \in \mathcal{U}_N$.
- Initialize ϕ as an empty set. Define Ω as the set of vehicles which receive more than one matching proposal. $\Omega = \emptyset$ at the beginning.
- Set $P_m = 0$ for any vehicle $V_m \in \mathcal{V}_M$.

Algorithm 1: Contract-Matching Algorithm.

```

1: Input:  $M, N, \{\theta_k\}, \{\lambda_k\}$ 
2: Output:  $\mathcal{C}^*, \phi$ 
Stage I: Contract-based Incentive Mechanism Design
3: Sort the types of vehicles based on (4);
4: Obtain the optimal contract  $\mathcal{C}^*$  by solving (20);
5: for  $V_m \in \mathcal{V}_M$  do
6:   Calculate the maximum amount of shared resource
      $\delta_m^{upper}$  based on (3).
7:   if  $\theta_k \leq \delta_m^{upper} < \theta_{k+1}$  then
8:      $V_m$  signs the contract item  $(\delta_k^*, \pi_k^*)$  with the base
       station, and shares its idle resource  $\delta_k^*$ .
9:   end if
10: end for
Stage II: Matching-based Task Assignment
11: Set  $\phi = \emptyset, \Omega = \emptyset, \Delta P_m = 0.1, P_m = 0$  and  $\mathcal{U}_{v_m} = \emptyset$  for
    each  $V_m \in \mathcal{V}_M$ ;
12: Every UE  $U_n \in \mathcal{U}_M$  builds its preference list  $F_n$  based
    on (30) and (31);
13: while  $\exists \phi(U_n) = \emptyset$  do
14:   for  $U_n \in \mathcal{U}_N$  do
15:     UE  $U_n$  proposes to its most preferred vehicle in its
       updated preference list  $F_n$ ;
16:   end for
17:   for  $V_m \in \mathcal{V}_M$  do
18:     if  $V_m$  receives more than one request then
19:        $\Omega = \Omega \cup \{V_m\}$ ;
20:       Add UEs which proposed to vehicle  $V_m$  to the
       set  $\mathcal{U}_{v_m}$ ;
21:     end if
22:   end for
23:   if  $\Omega = \emptyset$  then
24:     Match vehicles with UEs based on the proposals;
25:   else
26:     for  $V_m \in \Omega$  do
27:       Vehicle  $V_m$  increases its price  $P_m$  by  $\Delta P_m$ 
       based on (33);
28:       Every  $U_n \in \mathcal{U}_{v_m}$  updates its preference based
       on (30), and renews its proposals;
29:       Remove  $U_n$  from  $\mathcal{U}_{v_m}$  if it gives up  $V_m$ ;
30:       Remove  $V_m$  from  $\Omega$  if it receives only one
       proposal.
31:     end for
32:   end if
33: end while

```

Phase 2: Iterative Matching

Repeat the following process iteratively.

- If $\exists \phi(U_n) = \emptyset$, preform the propose rule for UEs.
 - Each UE $U_n \in \mathcal{U}_N$ proposes to its most preferred vehicle in its preference list F_n .
- If any vehicle $V_m \in \mathcal{V}_M$ receives only one proposal from a UE, then V_m will be directly matched with the UE which has proposed to it. Otherwise, add V_m into set Ω .
- If $\Omega \neq \emptyset$, perform the price rising rule for vehicles which received more than one matching proposal.

- Each vehicle $V_m \in \Omega$ increases its price by ΔP_m .
- Every UE which has proposed to V_m updates its preference towards V_m accordingly and renews its proposal.
- Remove the vehicles which receive only one proposal from Ω .

Until Every U_n has been matched with a vehicle, i.e., $\forall \phi(U_n) \neq \emptyset$, or there exists no available vehicle for the unmatched UEs.

Phase 3: Task Assignment Implementation

The UEs offload their tasks according to the matching results obtained in **Phase 2**. Assuming $\phi(U_n) = V_m$, UE U_n will send its task to vehicle V_m for processing. For those unmatched UEs in set Φ , their tasks will be processed by the base station. In the next slot, the base station updates the sets $\mathcal{U}_N$ and $\mathcal{V}_M$, and the task assignment process will return to **Phase 1**.

Remark 7: It is noted that any vehicle $V_m \in \mathcal{V}_M$, which cannot satisfy the delay constraint of U_n , will be directly removed from F_n despite of the preference.

D. Stability, Optimality, and Complexity Analysis

Definition 7. (Stability): A matching ϕ is stable if for any $U_n \in \mathcal{U}_N$, there does not exit a V_m such that $V_m \succ_{U_n} \phi(U_n)$.

Theorem 4: The pricing-based matching algorithm produces a stable matching within finite iterations.

Theorem 5: The obtained stable matching is weak Pareto optimal for UEs.

Proof: A similar proof can be found in our previous works [1], [12]. ■

Remark 8. (Computational Complexity): In the contract-based incentive mechanism design, the formulated optimization problem is a standard convex programming problem with M equality constraints and $2M + 1$ inequality constraints. The overall computation complexity is $\mathcal{O}(M)$.

In the process of the pricing-based matching, the complexity for each UE to acquire the preferences of all the vehicles and the complexity for sorting the obtained preferences are $\mathcal{O}(M)$ and $\mathcal{O}(M \log(M))$, respectively. Assuming the number of iterations required for resolving the conflict in the price rising process is ζ , i.e., the conflicting elements are matched within ζ iterations. Hence, the complexity of the matching process is $\mathcal{O}(N\zeta)$ ($N \geq M$) or $\mathcal{O}(M\zeta)$ ($M \geq N$).

The optimal matching result can be obtained by employing the exhaustive searching scheme. The total number of possible combinations is $(M^N N!)$. Thus, the exhaustive searching scheme has to examine every possible combination in order to find out the optimal matching result. The computation complexity is $\mathcal{O}(M^N N!)$.

VI. SIMULATION RESULTS

In this section, we validate the proposed scheme via simulations.

A. Contract Feasibility and Efficiency

A series of simulations are conducted to verify the feasibility and the efficiency of the contract-based incentive mechanism.

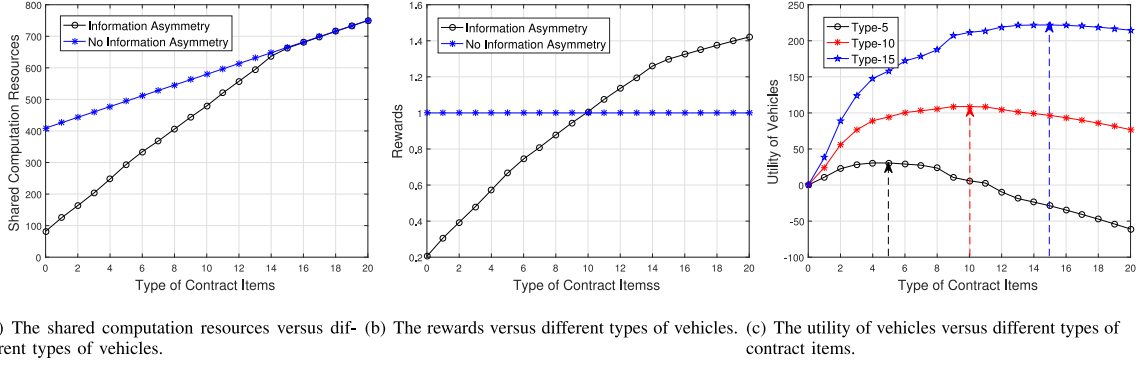


Fig. 2. Contract feasibility: (a) shared computation resources; (b) rewards; (c) utility of vehicles.

TABLE I
PARAMETERS

Parameter	Value
Number of vehicles	5 – 20
Number of UEs	6 – 25
Data size of UE's task	100 – 200 Mb
Computation size of UE's task	100 – 400 Mb
Delay constraint	0.1 – 2 s
Velocity of vehicles	2 – 20 m/s
Cell radius	1000 m
Radius of the UE's communication coverage	200 m
Computing resources of the base station	5 GHz
Transmission power of UEs	30 dBm
Bandwidth of UEs	20 MHz
Noise power	−114 dBm
Path loss exponent	−3.4

We consider a single cell with one base station, 20 vehicles, and 30 UEs. The number of vehicle types is equal to that of vehicles, and assume the vehicle types are following a Gaussian distribution. Simulation parameters are summarized in Table I. The proposed scheme is compared with the contract without information asymmetry [34] and the take-it-or-leave contract [23]. In the take-it-or-leave contract, the base station offers a uniform contract item which is designed based on a threshold type k_{th} . Then, vehicles with higher types, i.e., $k \geq k_{th}, \forall k \in \mathcal{K}$, will accept the contract, while vehicles with lower types, i.e., $k < k_{th}, \forall k \in \mathcal{K}$, will reject the contract.

Fig. 2(a) and Fig. 2(b) show the amount of shared computation resources and the rewards versus different vehicle types, respectively. As we can see, the computation resources that can be shared by vehicles and the corresponding reward increase monotonically with the vehicle type, which have been already demonstrated in Lemma 2. Furthermore, numerical results show that the contract without information asymmetry requires more resources from vehicles compared to that of information asymmetry. The reward provided for each vehicle is exactly 1 regardless of the vehicle type, which is consistent with Theorem 3.

Fig. 2(c) shows the utilities of type 5, type 10, and type 15 vehicles versus the different types of contract items. It is observed that each vehicle can maximize its utility if and only if it selects the specific contract item dedicated for its type. Furthermore, numerical results show that the utilities of vehicles also increase with the vehicle type, which agrees with the analysis summarized in Lemma 2.

Fig. 3(a) and Fig. 3(b) show the utility of the base station and the utility of vehicles versus the vehicle type. Numerical results

show that the asymmetric information actually protects the vehicles from being overexploited by the base station. With complete information, the base station is able to design a contract such that its utility is much larger compared to the utility achieved under the information asymmetry scenario. The performance gap increases monotonically with the vehicle type. Moreover, the contract enforces every vehicle's utility to be zero. The reason behind has been analyzed in the proof of Theorem 2. Therefore, information asymmetry is actually beneficial to the vehicles because the base station cannot overexploit a vehicle without knowing the complete information of its type.

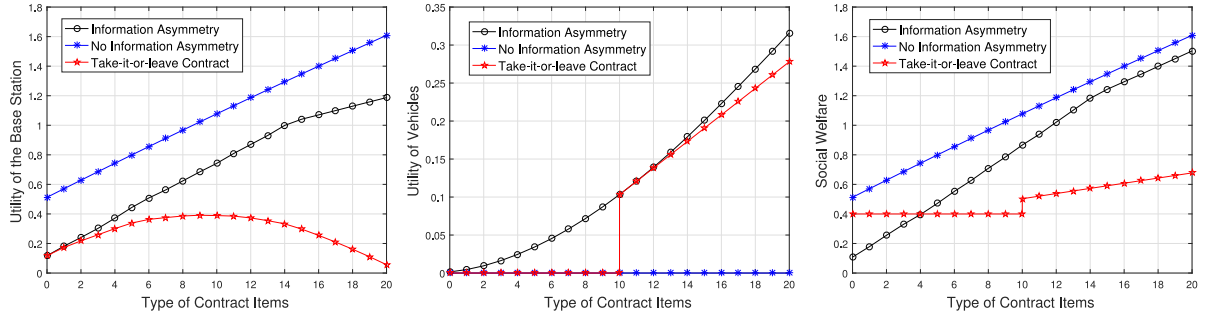
In the take-it-or-leave contract, any vehicle whose type satisfies $k < k_{th}, \forall k \in \mathcal{K}$ will reject the contract due to constraint C_4 . In this case, either the base station's utility or the vehicle's utility is zero. Only the vehicles with higher types, i.e., $k \geq k_{th}, \forall k \in \mathcal{K}$, can achieve nonzero utilities. However, since the take-it-or-leave contract is designed based on threshold type k_{th} , the gap between the utility of type k vehicle and that of the proposed scheme increases along with k . The reason is that the take-it-or-leave contract is not incentive compatible.

Fig. 3(c) shows the social welfare versus the vehicle type. Numerical results demonstrate that the social welfare achieved by the proposed contract is close to that of the contract without information asymmetry. The reason is that the utility of the base station obtained by exploiting the complete information leads to enormous utility loss of vehicles. On the other hand, the take-it-or-leave contract achieves the lowest social welfare.

B. Delay Performance

In simulations, we employ the constant-velocity model [10], [20], and the velocities of vehicles are generated randomly within the range [2, 20] m/s. To provide a relative comparison, the delay performances of different algorithms are normalized and converged to the range of [0, 1] by dividing the largest delay.

Fig. 4 shows the normalized network delay versus the number of matching iterations. Numerical results show that the proposed scheme can converge to a stable matching within a limited number of iterations. It is also observed that both the number of iterations required to reach convergence and the normalized network delay increase with the number of UEs. The reason is that the competition among UEs becomes more intense as the number of UEs increases. As a result, additional price-rising iterations are necessary to resolve the competition. Moreover, if



(a) The utility of the base station versus different types of vehicles. (b) The utilities of vehicles versus different types of vehicles. (c) The social welfare versus different types of vehicles.

Fig. 3. Contract efficiency: (a) utility of the base station; (b) utilities of vehicles; (c) social welfare.

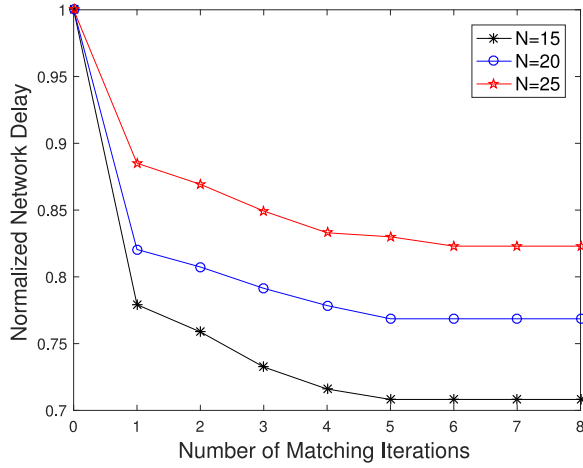


Fig. 4. Normalized delay of UEs vary with the number of iterations.

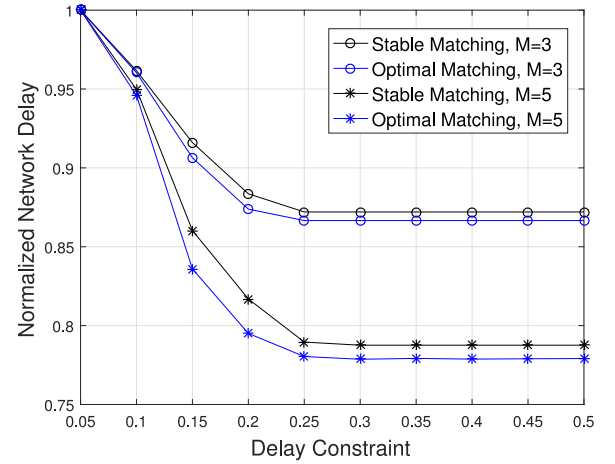


Fig. 6. Normalized delay versus the delay constraint τ_n .

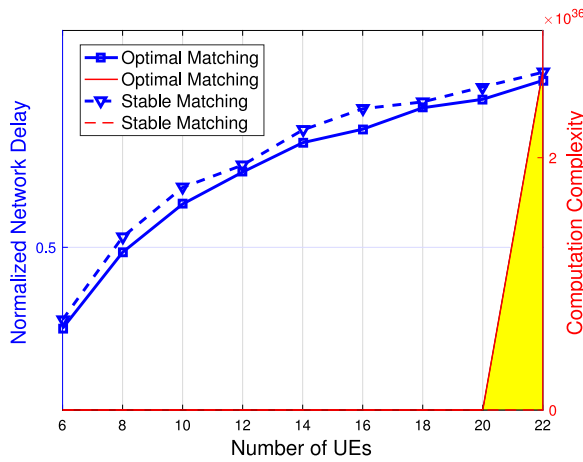


Fig. 5. Normalized delay and computation complexity versus the number of UEs.

the number of UEs is much larger than that of vehicles, a larger amount of tasks have to be processed by the overloaded base station, which significantly increases the network delay.

Fig. 5 shows the normalized network delay versus the number of UEs. Numerical results demonstrate that the network delay is inversely related to the number of vehicles and positively related to the number of UEs, which is consistent with the results of Fig. 4. Furthermore, the proposed scheme is able to achieve a

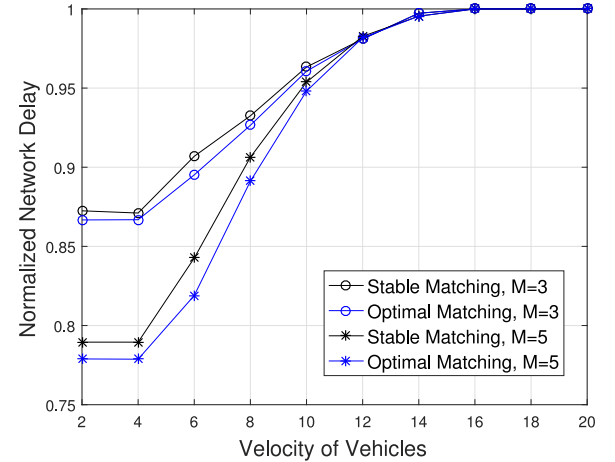


Fig. 7. Normalized delay versus the velocity of vehicles.

network delay that is close to that of the optimal exhaustive searching algorithm but with a much lower complexity.

Fig. 6 shows the normalized network delay versus the total delay constraint of the computation task. As the total delay constraint increases, the number of eligible vehicles also increases accordingly, and more and more tasks can be offloaded to the vehicles rather than being processed by the overloaded base station with limited computation resources. This will dramatically reduce the network delay since the under-utilized resources

of vehicles have been well exploited. Numerical results also demonstrate that the proposed scheme can achieve close-to-optimal performance under all the investigated scenarios.

Fig. 7 shows the normalized network delay versus the vehicle velocity. Numerical results demonstrate that the network delay increases with the vehicle velocity. The reason is that the number of eligible vehicles decreases along with the vehicle velocity due to the stringent constraint of transmission delay. As a result, it is less likely for a UE to be matched with a satisfactory vehicle, and the corresponding task can only be processed by the overloaded base station. Hence, the proposed scheme is more suitable for hot spots where there exist a large number of parked vehicles or the vehicles move slowly due to traffic congestion.

VII. CONCLUSION

In this paper, we investigated the computation resource allocation and task assignment problem in VFC from a contract-matching integration perspective. A contract-based incentive mechanism was proposed to motivate vehicles to share their resources, and a pricing-based stable matching algorithm was developed to address the task assignment problem. Numerical results demonstrate that the proposed incentive mechanism achieves a social welfare that is close to the optimal performance without information asymmetry, while the proposed task assignment scheme is able to achieve a network delay that is close to that of the optimal exhaustive searching algorithm but with a much lower complexity.

In future works, we will investigate the more complicated scenario where the precise knowledge of channel and vehicle states is unknown, and study how to combine machine learning-based approaches to optimize the long-term delay performance.

Furthermore, the wide-scale deployment of VFC faces numerous security and forensic challenges. Different from cloud computing where the servers are deployed, operated, and maintained by specialized service providers, the fog nodes are generally deployed and maintained by third-party developers or even users. Therefore, to guarantee the reliable operation of VFC, several security mechanisms including confidentiality, integrity, authentication, access control, and forensics, etc., are required [41]. However, some existing security solutions developed for cloud computing may not scale well in VFC [42]. Particularly, the heterogeneous and distributed nature of fog nodes poses new threats on authentication, access control, and resilience [42]. In summary, the research on the security aspect of VFC is still in the infancy stage, which requires further investigation and examination.

REFERENCES

- [1] Z. Zhou, C. Gao, C. Xu, Y. Zhang, S. Mumtaz, and J. Rodriguez, "Social big-data-based content dissemination in Internet of Vehicles," *IEEE Trans. Ind. Inform.*, vol. 14, no. 2, pp. 768–777, Feb. 2018.
- [2] N. Wang, B. Varghese, M. Matthaiou, and D. S. Nikolopoulos, "ENORM: A framework for edge node resource management," *IEEE Trans. Serv. Comput.*, to be published, doi: 10.1109/TSC.2017.2753775.
- [3] Z. Zhou, H. Liao, B. Gu, K. Huq, S. Mumtaz, and J. Rodriguez, "Robust mobile crowd sensing: When deep learning meets edge computing," *IEEE Netw.*, vol. 32, no. 4, pp. 54–60, Aug. 2018.
- [4] H. Guo, J. Liu, J. Zhang, W. Sun, and N. Kato, "Mobile-edge computation offloading for ultra-dense IoT networks," *IEEE Internet Things J.*, vol. 5, no. 6, pp. 4977–4988, Dec. 2018.
- [5] Y. Cui, W. He, C. Ni, C. Guo, and Z. Liu, "Energy-efficient resource allocation for cache-assisted mobile edge computing," in *Proc. IEEE Local Comput. Netw.*, Singapore, Oct. 2017, pp. 640–648.
- [6] J. Guo, Z. Song, Y. Cui, Z. Liu, and Y. Ji, "Energy-efficient resource allocation for multi-user mobile edge computing," in *Proc. IEEE Global Commun. Conf.*, Singapore, Dec. 2017, pp. 1–7.
- [7] Y. Lai, F. Yang, L. Zhang, and Z. Lin, "Distributed public vehicle system based on fog nodes and vehicular sensing," *IEEE Access*, vol. 6, pp. 22011–22024, 2018.
- [8] Q. Yuan, H. Zhou, J. Li, Z. Liu, F. Yang, and X. Shen, "Toward efficient content delivery for automated driving services: An edge computing solution," *IEEE Netw.*, vol. 32, no. 1, pp. 80–86, Jan. 2018.
- [9] H. Zhang, Q. Zhang, and X. Du, "Toward vehicle-assisted cloud computing for smartphones," *IEEE Trans. Veh. Technol.*, vol. 64, no. 12, pp. 5610–5618, Sep. 2015.
- [10] X. Hou, Y. Li, M. Chen, D. Wu, D. Jin, and S. Chen, "Vehicular fog computing: A viewpoint of vehicles as the infrastructures," *IEEE Trans. Veh. Technol.*, vol. 65, no. 6, pp. 3860–3873, Jun. 2016.
- [11] H. Peng et al., "Resource allocation for cellular-based inter-vehicle communications in autonomous platoons," *IEEE Trans. Veh. Technol.*, vol. 66, no. 12, pp. 11249–11263, Dec. 2017.
- [12] Z. Zhou, K. Ota, M. Dong, and C. Xu, "Energy-efficient matching for resource allocation in D2D enabled cellular networks," *IEEE Trans. Veh. Technol.*, vol. 66, no. 6, pp. 5256–5268, Jun. 2017.
- [13] W. Zhang, Z. Zhang, and H. Chao, "Cooperative fog computing for dealing with big data in the Internet of Vehicles: Architecture and hierarchical resource management," *IEEE Commun. Mag.*, vol. 55, no. 12, pp. 60–67, Dec. 2017.
- [14] J. Feng, Z. Liu, C. Wu, and Y. Ji, "AVE: Autonomous vehicular edge computing framework with ACO-based scheduling," *IEEE Trans. Veh. Technol.*, vol. 66, no. 12, pp. 10660–10675, Dec. 2017.
- [15] M. Satyanarayanan, "Edge computing for situational awareness," in *Proc. IEEE Int. Symp. Local Metrop. Area Netw.*, Osaka, Japan, Jun. 2017, pp. 1–6.
- [16] Y. Xiao and C. Zhu, "Vehicular fog computing: Vision and challenges," in *Proc. IEEE Int. Conf. Pervasive Comput. Commun. Workshops*, Kona, HI, USA, Mar. 2017, pp. 6–9.
- [17] C. Zhu, G. Pastor, Y. Xiao, Y. Li, and A. Ylae-Jaaski, "Fog following me: Latency and quality balanced task allocation in vehicular fog computing," in *Proc. IEEE Annu. Int. Conf. Sens., Commun. Netw.*, Hong Kong, Jun. 2018, pp. 1–9.
- [18] N. C. Luong, P. Wang, D. Niyato, Y. Wen, and Z. Han, "Resource management in cloud networking using economic analysis and pricing models: A survey," *IEEE Commun. Surv. Tut.*, vol. 19, no. 2, pp. 954–1001, Apr.–Jun. 2017.
- [19] Y. Liu, C. Xu, Y. Zhan, Z. Liu, J. Guan, and H. Zhang, "Incentive mechanism for computation offloading using edge computing: A Stackelberg game approach," *Comput. Netw.*, vol. 129, no. 2, pp. 399–409, Dec. 2017.
- [20] Z. Su, Q. Xu, Y. Hui, M. Wen, and S. Guo, "A game theoretic approach to parked vehicle assisted content delivery in vehicular ad-hoc networks," *IEEE Trans. Veh. Technol.*, vol. 66, no. 7, pp. 6461–6474, Nov. 2017.
- [21] J. Du, E. Gelenbe, C. Jiang, H. Zhang, and Y. Ren, "Contract design for traffic offloading and resource allocation in heterogeneous ultra-dense networks," *IEEE J. Sel. Areas Commun.*, vol. 35, no. 11, pp. 2457–2467, Nov. 2017.
- [22] L. Xu, C. Jiang, Y. Shen, T. Q. S. Quek, Z. Han, and Y. Ren, "Energy efficient D2D communications: A perspective of mechanism design," *IEEE Trans. Wireless Commun.*, vol. 15, no. 11, pp. 7272–7285, Nov. 2016.
- [23] L. Duan, L. Gao, and J. Huang, "Cooperative spectrum sharing: A contract-based approach," *IEEE Trans. Mobile Comput.*, vol. 13, no. 1, pp. 174–187, Jan. 2014.
- [24] Y. Zhang, M. Pan, L. Song, Z. Dawy, and Z. Han, "A survey of contract theory-based incentive mechanism design in wireless networks," *IEEE Wireless Commun.*, vol. 24, no. 3, pp. 80–85, Jun. 2017.
- [25] T. Liu, J. Li, F. Shu, M. Tao, W. Chen, and Z. Han, "Design of contract-based trading mechanism for a small-cell caching system," *IEEE Trans. Wireless Commun.*, vol. 16, no. 10, pp. 6602–6617, Oct. 2017.
- [26] L. Duan, T. Kubo, K. Sugiyama, J. Huang, T. Hasegawa, and J. Walrand, "Motivating smartphone collaboration in data acquisition and distributed computing," *IEEE Trans. Mobile Comput.*, vol. 13, no. 10, pp. 2320–2333, Oct. 2014.
- [27] Z. Zhou, J. Feng, B. Gu, B. Ai, S. Mumtaz, and J. Rodriguez, "When mobile crowd sensing meets UAV: Energy-efficient task assignment and route planning," *IEEE Trans. Commun.*, vol. 66, no. 11, pp. 5526–5538, Jul. 2018.

- [28] Z. Zhou, H. Yu, C. Xu, Y. Zhang, S. Mumtaz, and J. Rodriguez, "Dependable content distribution in D2D-based cooperative vehicular networks: A big data-integrated coalition game approach," *IEEE Trans. Intell. Transp. Syst.*, vol. 19, no. 3, pp. 953–964, Jan. 2018.
- [29] H. Shah-Mansouri and V. W. S. Wong, "Hierarchical fog-cloud computing for IoT systems: A computation offloading game," *IEEE Internet Things J.*, vol. 5, no. 4, pp. 3246–3257, Aug. 2018.
- [30] K. Zhang, Y. Mao, S. Leng, Y. He, and Y. Zhang, "Mobile-edge computing for vehicular networks: A promising network paradigm with predictive offloading," *IEEE Veh. Technol. Mag.*, vol. 12, no. 2, pp. 36–44, Apr. 2017.
- [31] H. Zhang, Y. Xiao, S. Bu, D. Niyato, F. R. Yu, and Z. Han, "Computing resource allocation in three-tier IoT fog networks: A joint optimization approach combining Stackelberg game and matching," *IEEE Internet Things J.*, vol. 4, no. 5, pp. 1204–1215, Mar. 2017.
- [32] D. T. Nguyen, L. Le, and V. Bhargava, "Price-based resource allocation for edge computing: A market equilibrium approach," *IEEE Trans. Cloud Comput.*, to be published, doi: [10.1109/TCC.2018.2844379](https://doi.org/10.1109/TCC.2018.2844379).
- [33] A. Kiani and N. Ansari, "Toward hierarchical mobile edge computing: An auction-based profit maximization approach," *IEEE Internet Things J.*, vol. 4, no. 6, pp. 2082–2091, Sep. 2017.
- [34] Y. Zhang, L. Song, W. Saad, Z. Dawy, and Z. Han, "Contract-based incentive mechanisms for device-to-device communications in cellular networks," *IEEE J. Sel. Areas Commun.*, vol. 33, no. 10, pp. 2144–2155, Oct. 2015.
- [35] S. Zhao, Y. Chen, and J. Farrell, "High-precision vehicle navigation in urban environments using an MEM's IMU and single-frequency GPS receiver," *IEEE Wireless Commun.*, vol. 17, no. 10, pp. 2854–2867, Apr. 2016.
- [36] F. Malandrino, C. Casetti, C. Chiasserini, and M. Fiore, "Content download in vehicular networks in presence of noisy mobility prediction," *IEEE Trans. Mobile Comput.*, vol. 13, no. 5, pp. 1007–1021, Oct. 2013.
- [37] Y. Li, D. Jin, P. Hui, Z. Wang, and S. Chen, "Limits of predictability for large-scale urban vehicular mobility," *IEEE Trans. Intell. Transp. Syst.*, vol. 15, no. 6, pp. 2671–2682, Dec. 2014.
- [38] B. Kim and K. Yi, "Probabilistic and holistic prediction of vehicle states using sensor fusion for application to integrated vehicle safety systems," *IEEE Trans. Intell. Transp. Syst.*, vol. 15, no. 5, pp. 2178–2190, Oct. 2014.
- [39] J. Kim and D. Kum, "Threat prediction algorithm based on local path candidates and surrounding vehicle trajectory predictions for automated driving vehicles," in *Proc. IEEE Intell. Veh. Symp.*, Seoul, South Korea, Aug. 2015, pp. 1220–1225.
- [40] A. Roth and M. Sotomayor, *Two-Sided Matching: A Study in Game-Theoretic Modeling and Analysis*. Cambridge, U.K.: Cambridge Univ. Press, 1992.
- [41] S. N. Shirazi, A. Gougilidis, A. Farshad, and D. Hutchison, "The extended cloud: Review and analysis of mobile edge computing and fog from a security and resilience perspective," *IEEE J. Sel. Areas Commun.*, vol. 35, no. 11, pp. 2586–2595, Oct. 2017.
- [42] C. Huang, R. Lu, and K. Choo, "Vehicular fog computing: Architecture, use case, and security and forensic challenges," *IEEE Commun. Mag.*, vol. 55, no. 11, pp. 105–111, Nov. 2017.



Zhenyu Zhou (M'11–SM'17) received the M.E. and Ph.D. degrees from Waseda University, Tokyo, Japan, in 2008 and 2011, respectively. From April 2012 to March 2013, he was the Chief Researcher with the Department of Technology, KDDI, Tokyo, Japan. Since March 2013, he has been an Associate Professor with the School of Electrical and Electronic Engineering, North China Electric Power University, Beijing, China. His research interests include green communications, vehicular communications, and smart grid communications. He was an Associate

Editor for IEEE ACCESS and EURASIP JOURNAL ON WIRELESS COMMUNICATIONS AND NETWORKING, and a Guest Editor for the IEEE COMMUNICATIONS MAGAZINE and the IEEE TRANSACTIONS ON EMERGING TELECOMMUNICATIONS TECHNOLOGIES. He was also the Workshop Co-Chair for IEEE Globecom 2018 and IEEE ISADS 2015, and a TPC member for IEEE Globecom, IEEE CCNC, IEEE ICC, IEEE APCC, IEEE VTC, IEEE Africon, etc. He is a voting member of IEEE Standard Association P1932.1 Working Group. He was the recipient of the IEEE Vehicular Technology Society's Young Researcher Encouragement Award in 2009, the Beijing Outstanding Young Talent Award in 2016, the IET Premium Award in 2017, the IEEE ComSoc Green Communications and Computing Technical Committee 2017 Best Paper Award, the IEEE Globecom 2018 Best Paper Award, and the IEEE ComSoc Green Communications and Computing Technical Committee 2018 Best Paper Award.



Pengju Liu is currently working toward the bachelor's degree with North China Electric Power University, Beijing, China. His research interests include resource allocation and energy management in VEC.



Junhao Feng is currently working toward the M.S. degree with North China Electric Power University, Beijing, China. His research interests include resource allocation, interference management, and energy management in D2D communications. He was the recipient of the IEEE Globecom 2018 Best Paper Award, the IEEE ComSoc Green Communications, and the Computing Technical Committee 2018 Best Paper Award.



Yan Zhang (M'05–SM'10) received the Ph.D. degree from the School of Electrical and Electronics Engineering, Nanyang Technological University, Singapore. He is currently a Full Professor with the Department of Informatics, University of Oslo, Oslo, Norway. His current research interests include next-generation wireless networks leading to 5G, green and secure cyber-physical systems, such as smart grid, healthcare, and transport. He is an IEEE Vehicular Technology Society (VTS) Distinguished

Lecturer. He is also a Senior Member of the IEEE ComSoc, the IEEE CS, the IEEE PES, and the IEEE VTS. He is a Fellow of the IET. He is an Associate Technical Editor for the IEEE COMMUNICATIONS MAGAZINE, an Editor for the IEEE NETWORK MAGAZINE, the IEEE TRANSACTIONS ON GREEN COMMUNICATIONS AND NETWORKING, the IEEE COMMUNICATIONS SURVEYS AND TUTORIALS, and the IEEE INTERNET OF THINGS JOURNAL, and an Associate Editor for the IEEE ACCESS. He is the Chair in a number of conferences, including the IEEE GLOBECOM 2017, the IEEE PIMRC 2016, the IEEE CloudCom 2016, the IEEE ICC 2016, the IEEE CCNC 2016, the WCSP 2016, the IEEE SmartGridComm 2015, and the IEEE CloudCom 2015. He is a TPC member for numerous international conferences, including the IEEE INFOCOM, the IEEE ICC, the IEEE GLOBECOM, and the IEEE WCNC.



Shahid Mumtaz (SM'16) received the M.Sc. degree from the Blekinge Institute of Technology, Karlskrona, Sweden, and the Ph.D. degree from the University of Aveiro, Aveiro, Portugal. He is currently a Senior Research Engineer with the Instituto de Telecomunicações, Aveiro, Portugal, where he is involved in EU funded projects. His research interests include MIMO techniques, multi-hop relaying communication, cooperative techniques, cognitive radios, game theory, energy efficient framework for 4G, position-information-assisted communication, and joint PHY and MAC layer optimization in LTE standard. He has authored several conferences, journals, and books publications.



Jonathan Rodriguez (M'04–SM'13) received the master's and Ph.D. degrees in electronic and electrical engineering from the University of Surrey, Guildford, U.K., in 1998 and 2004, respectively. In 2005, he became a Researcher with the Instituto de Telecomunicações, and in 2008, he became the Senior Researcher, where he established the 4TELL Research Group targeting the next-generation mobile networks with key interests on energy efficient design, cooperative strategies, security, and electronic circuit design. He was a Project Coordinator for major

international research projects (Eureka LOOP, FP7 C2POWER), while acting as the Technical Manager for FP7 COGEU and FP7 SALUS. He is currently leading the H2020-ETN SECRET project. Since 2009, he has been an Assistant Professor with the Universidade de Aveiro, Aveiro, Portugal, and was granted as an Associate Professor in 2015. His professional affiliations include Chartered Engineer (CEng) since 2013, and fellow of the IET (2015).